

Sparse Table

1 Introduction

A Sparse Table is a data structure used to answer **static range queries** efficiently. It is most commonly used for:

- Range Minimum Query (RMQ)
- Range Maximum Query

It supports:

- **Preprocessing:** $O(n \log n)$
- **Query time:** $O(1)$

Sparse Table works only for **immutable (static) arrays**.

2 Basic Idea

Sparse Table stores answers for intervals of length 2^k .

$$ST[i][k] = \text{minimum value in range } [i, i + 2^k - 1]$$

Any query range $[L, R]$ can be answered using two overlapping intervals.

3 Example Array

$$A = [7, 2, 3, 0, 5, 10, 3, 12, 18]$$

Number of elements:

$$n = 9$$

4 Log Table Construction

To compute powers of two efficiently, we precompute logarithms:

$$\log_2 = [0, 0, 1, 1, 2, 2, 2, 2, 3]$$

Formula:

$$\log[i] = \log[i/2] + 1$$

5 Sparse Table Construction

5.1 Level $k = 0$ (Intervals of length $2^0 = 1$)

$$ST[i][0] = A[i]$$

i	$ST[i][0]$
0	7
1	2
2	3
3	0
4	5
5	10
6	3
7	12
8	18

5.2 Level $k = 1$ (Intervals of length $2^1 = 2$)

$$ST[i][1] = \min(ST[i][0], ST[i+1][0])$$

i	$ST[i][1]$
0	2
1	2
2	0
3	0
4	5
5	3
6	3
7	12

5.3 Level $k = 2$ (Intervals of length $2^2 = 4$)

$$ST[i][2] = \min(ST[i][1], ST[i+2][1])$$

i	$ST[i][2]$
0	0
1	0
2	0
3	0
4	3
5	3

5.4 Level $k = 3$ (Intervals of length $2^3 = 8$)

$$ST[i][3] = \min(ST[i][2], ST[i + 4][2])$$

i	$ST[i][3]$
0	0
1	0

6 Range Minimum Query

To answer $RMQ(L, R)$:

$$j = \log_2(R - L + 1)$$

$$RMQ(L, R) = \min(ST[L][j], ST[R - 2^j + 1][j])$$

7 Worked Query Example

Query: $RMQ(1, 6)$

$$\text{Length} = 6 \Rightarrow j = \lfloor \log_2 6 \rfloor = 2$$

$$RMQ(1, 6) = \min(ST[1][2], ST[3][2])$$

$$= \min(0, 0) = 0$$

8 Algorithms

8.1 Build Sparse Table

Algorithm 1 Build Sparse Table

Require: Array $A[0 \dots n - 1]$

Ensure: Sparse Table ST

```

1: for  $i = 0$  to  $n - 1$  do
2:    $ST[i][0] \leftarrow A[i]$ 
3: end for
4: for  $k = 1$  to  $\lfloor \log_2 n \rfloor$  do
5:   for  $i = 0$  while  $i + 2^k \leq n$  do
6:      $ST[i][k] \leftarrow \min(ST[i][k - 1], ST[i + 2^{k-1}][k - 1])$ 
7:   end for
8: end for
```

8.2 Query

Algorithm 2 Range Minimum Query

Require: L, R

- 1: $j \leftarrow \lfloor \log_2(R - L + 1) \rfloor$
 - 2: return $\min(ST[L][j], ST[R - 2^j + 1][j])$
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9 Time and Space Complexity

- Preprocessing Time: $O(n \log n)$
- Query Time: $O(1)$
- Space Complexity: $O(n \log n)$

10 Advantages and Limitations

Advantages

- Extremely fast queries
- Simple implementation

Limitations

- No updates allowed
- Higher memory usage

11 Conclusion

Sparse Table is a powerful technique for answering static range queries efficiently. It is ideal when the array does not change and multiple queries must be answered quickly.